

4. Literature Review

4.1 Introduction

The rapid acceleration of the global digital economy has established information communication technologies (ICTs) as indispensable engines for enterprise survival, operational agility, and sustainable competitive advantage. Within this paradigm shift, cloud computing has emerged as a disruptive technological megatrend, fundamentally redefining how modern enterprises architect, deploy, and manage their computing infrastructure. By decoupling enterprise software applications and storage repositories from capital-intensive physical hardware, cloud computing democratises access to sophisticated computational resources. This democratisation provides resource-constrained organisations with unprecedented levels of scalability and processing capabilities.

For Small and Medium Enterprises (SMEs) in South Africa, which operate within a highly volatile economic landscape marked by structural inequalities and severe infrastructural deficits, the strategic adoption of cloud computing is no longer a luxury but a survival imperative. Despite these well-documented advantages, empirical evidence demonstrates that South African SMEs exhibit a paradoxical lag in cloud adoption. They continue to operate on archaic, fragmented on-premises systems or basic digital tools that severely constrain their growth.

To diagnose and resolve the multi-dimensional bottlenecks responsible for this low adoption rate, a comprehensive, systematic synthesis of existing scholarship is required. This literature review provides a rigorous, critical evaluation of the theoretical constructs, empirical debates, and contextual anomalies that govern cloud computing adoption. Rather than presenting a disjointed summary of prior studies, this review utilises a narrative synthesis approach structured around the Technology-Organisation-Environment (TOE) framework.

The TOE framework provides an epistemological lens capable of dissecting the interplay between internal technical configurations, firm-level structural dynamics, and external systemic realities specific to South Africa, such as legislative mandates and electricity crises. Ultimately, this chapter maps the structural architecture of current global and localised research, exposes major contradictions within contemporary information systems scholarship, and establishes the precise empirical gap that justifies this study.

4.2 Cloud Computing: Conceptual Background

4.2.1 Definition of Cloud Computing

Defining cloud computing within academic discourse requires moving beyond commercial marketing rhetoric to examine its architectural and operational principles. The most widely accepted baseline definition in information systems research is formulated by Mell and Grance (2011:2) for the National Institute of Standards and Technology (NIST). They define cloud computing as:

"...a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction".

This definition emphasises five essential characteristics: on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.

While the NIST model serves as a standard baseline, subsequent scholars have expanded its technical focus to incorporate socio-technical and economic perspectives. For instance, Jayeola *et al.* (2022:228) criticise purely technical definitions for failing to account for the utility-computing business model, arguing that cloud computing is best understood as an economic transition from capital expenditure (CapEx) to operational expenditure (OpEx). Conversely, Mkhize *et al.* (2025:online) conceptualise cloud computing as an ecosystemic capability that shifts the locus of computing control away from the enterprise to third-party providers. This shift introduces complex institutional dependencies and risks that are not fully captured by the NIST definition.

The common element across these definitions is the elimination of physical computing constraints. However, a key difference lies in their emphasis: technical definitions focus on provisioning velocity, whereas business-centric definitions focus on financial restructuring. For South African SMEs, this conceptual duality is critical; cloud computing must be evaluated not only as a software delivery mechanism but as a strategic mechanism for resource conservation.

4.2.2 Service Models

The operational architecture of cloud computing is divided into three core service models: Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS). Each model creates a distinct boundary of technical responsibility between the enterprise and the cloud provider, offering different benefits and constraints for SMEs.

Software as a Service (SaaS)

SaaS represents the highest tier of abstraction, where end-user applications are delivered entirely over the internet, bypassing the need for local installation or maintenance. Empirical research by Ahmad, Mahomed and Hashim (2025:3) reveals that SaaS is the most widely adopted model among SMEs due to its low technical barriers to entry. Applications such as cloud-hosted accounting tools, CRM systems, and collaborative communication tools offer instant utility.

However, Mkhize *et al.* (2025:online) note that SaaS introduces limitations regarding customisation and vendor lock-in. SMEs adopting SaaS must align their business processes with the standardised configurations of the vendor, which can erode unique operational advantages.

Platform as a Service (PaaS)

PaaS occupies the middle tier, providing a virtualised development environment, operating systems, and database management engines that allow developers to build, test, and deploy

bespoke software applications. The primary advantage of PaaS is that it insulates firms from the complex realities of underlying server and operating system management. Nevertheless, as Yaseen *et al.* (2023:320) point out, PaaS assumes a baseline level of internal software engineering capability. For the vast majority of South African micro and small enterprises that lack dedicated IT development teams, PaaS remains underutilised and largely irrelevant to daily operations.

Infrastructure as a Service (IaaS)

IaaS represents the foundational tier, delivering raw computing power, virtual storage, firewalls, and network configurations on an on-demand basis. IaaS offers maximum operational flexibility, allowing firms to scale their infrastructure dynamically.

However, current literature highlights a severe trade-off: IaaS exhibits the highest configuration complexity, acting as a significant inhibitor for smaller firms (Yaseen *et al.*, 2023:320). It demands advanced network administration skills to secure and manage the virtualised assets effectively. Consequently, while large corporations leverage IaaS to achieve massive infrastructural cost savings, SMEs often find the configuration requirements an insurmountable barrier, exposing them to catastrophic security vulnerabilities if misconfigured.

4.2.3 Deployment Models

The strategic utility of cloud computing is further governed by its deployment topologies, classified into public, private, and hybrid configurations. These models dictate where data resides, who manages the physical hardware, and the multi-tenancy status of the environment.

Public Cloud

The public cloud operates on a shared, multi-tenant architecture where infrastructure is owned by global hyperscalers and accessed over the public internet. The primary benefit for South African SMEs is affordability, as it minimizes capital costs through shared infrastructure. However, Jayeola *et al.* (2022:228) caution that public clouds offer the lowest level of administrative control and raise significant compliance concerns under rigid data privacy regimes.

Private Cloud

In contrast, the private cloud features a single-tenant environment dedicated exclusively to one organisation, hosted either on-premises or by a specialised local provider. This architecture provides high levels of security, isolation, and control.

However, McKinsey and Company (2024:online) note that the exorbitant capital requirements needed to establish and maintain private clouds make them economically unviable for resource-constrained SMEs, restricting their use to highly capitalised institutions.

Hybrid Cloud

The hybrid cloud model seeks to bridge this divide by orchestrating workloads across both public and private environments. This approach allows an enterprise to store sensitive consumer data in a secure private repository while running resource-intensive application

processes within the scalable public cloud. Ahmad *et al.* (2025:3) argue that the hybrid cloud represents the most practical and resilient compromise for contemporary SMEs. This is particularly true in South Africa, where hybrid configurations can provide localised, offline-capable application continuity during periods of widespread power outages. Despite these strategic benefits, McKinsey and Company (2024:online) show that awareness and technical understanding of hybrid structures remain low across the African SME sector. This knowledge gap leaves many small businesses trapped in an all-or-nothing approach to cloud migration.

4.3 The Economic and Strategic Importance of SMEs in South Africa

To appreciate the urgency of accelerating cloud computing adoption, one must analyze the role that Small and Medium Enterprises play within the South African socioeconomic framework. It is widely agreed across economic and development literature that SMEs serve as the primary engine for employment creation, poverty alleviation, and gross domestic product (GDP) growth within emerging markets. According to Ngo Ndjama and Van der Westhuizen (2024:608), both formal and informal SMEs account for more than 90 percent of all operating businesses in South Africa. They employ approximately 60 percent of the national workforce and contribute around 40 percent of the country’s total GDP. Furthermore, the South African government’s National Development Plan (NDP 2030) projects that by the year 2030, small businesses must generate 90 percent of the 11 million new jobs required to stabilise the structural unemployment crisis. Despite this reliance on the small business sector, South African SMEs operate under severe structural pressures. FinMark Trust (2024:online) reports that a lack of access to digital tools, coupled with low technological adoption rates, threatens the survival of these enterprises. When comparing digitally transformed enterprises with technologically excluded SMEs, contemporary literature reveals clear disparities. Small businesses that rely on legacy, manual, spreadsheet-based data management cannot compete with agile, digitally enabled rivals. They suffer from higher operational overheads, poor customer acquisition rates, and limited access to cross-border supply chains.

Economic Indicator	SME Contribution / Target	Contextual Realities
Total Business Share	>90% of all formal & informal enterprises	Operates within deep structural inequalities
Current Employment	~60% of the active workforce	Threatened by low market competitiveness
GDP Contribution	~40% of national GDP	NDP target requires increase to 60-80%

NDP 2030 Job Target	90% of 11 million projected jobs	Dependent on bridging the digital divide
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Consequently, low digital adoption stalls individual business growth while undercutting national targets. Without integrating cloud platforms that offer enterprise-grade capabilities at scalable prices, South African SMEs risk further marginalisation within an increasingly digital marketplace. This dynamic could hinder economic recovery and worsen the national unemployment rate, which exceeded 33 percent in 2024.

4.4 The Technology–Organisation–Environment (TOE) Framework

4.4.1 Origins and Theoretical Foundations

To understand the factors driving or stalling technology adoption within an enterprise, researchers must use firm-level theoretical lenses rather than individual-level acceptance models. Models like the Technology Acceptance Model (TAM) or the Unified Theory of Acceptance and Use of Technology (UTAUT) focus on individual user perceptions and psychology. As a result, they fail to account for corporate governance, financial constraints, and institutional pressures.

To address these limitations, Tornatzky and Fleischer (1990:152) developed the Technology–Organisation–Environment (TOE) framework. This framework states that an enterprise's innovation adoption path is shaped by three distinct, interacting contexts:

- **The Technological Context:** Refers to the internal and external technologies available to the firm, evaluating their characteristics, compatibility, and complexity.
- **The Organisational Context:** Defined by the internal characteristics of the enterprise, including its size, formalisation, communication mechanisms, human capital skills, and resource constraints.
- **The Environmental Context:** Encompasses the external arena in which the firm conducts business, including industry structure, competitive pressures, government regulatory regimes, and macroeconomic infrastructure stability.

The theoretical strength of the TOE framework lies in its structural flexibility and comprehensive nature. Rather than treating technology implementation as an isolated event, TOE views it as a socio-technical process embedded within changing institutional frameworks. This comprehensive scope makes it well-suited for researching SME cloud adoption in developing economies. It allows researchers to map the technical attributes of cloud systems alongside internal firm resource limitations and external environmental challenges, such as unreliable utilities and regulatory mandates.

4.4.2 Application of TOE in Previous Studies

The academic literature shows that the TOE framework has been validated across various international and developing-country contexts. For instance, Santos *et al.* (2023:online) applied

the TOE framework to analyze Micro, Small, and Medium Enterprises (MSMEs) in Portugal, finding that combining these dimensions significantly improved firm performance post-adoption.

In a developing-economy context, Abu Bakar *et al.* (2024:556) used the framework to examine Malaysian SMEs, showing that environmental competitive pressure was a strong driver of technology adoption intent. Similarly, Jayeola *et al.* (2022:234) conducted a global systematic review and confirmed that TOE is the most frequently used framework for evaluating organisational cloud computing trends.

However, a critical review of these prior applications reveals distinct theoretical and geographical limitations. Many international studies treat the environmental context as a stable variable, focusing primarily on vendor support or market competition. This assumption does not hold in sub-Saharan Africa, where environmental factors include severe infrastructural challenges, like persistent rolling blackouts (load shedding), and strict data sovereignty laws. Scholars like Adane (2021:online) argue that while standard implementations of the TOE framework capture broad organisational realities, they often fail to capture the localised nuances of developing nations. Therefore, while the TOE framework provides an excellent structural foundation, it must be adapted to account for the unique regulatory and infrastructural realities faced by South African SMEs.

4.5 Technological Factors Influencing Cloud Adoption

Relative Advantage

Relative advantage denotes the degree to which an enterprise perceives cloud computing as superior to its existing legacy, on-premises infrastructure. Across both global and localised literature, relative advantage is widely recognised as a primary positive predictor of cloud adoption intent.

Ilahi *et al.* (2024:57) demonstrate that when SME managers recognise that cloud configurations lower operational costs and improve collaboration, adoption intent rises.

In South Africa, this advantage often focuses on converting high capital expenditures into scalable operational models. However, Mokwena and Sithole (2024:online) present a counter-narrative from their empirical study in the Polokwane Municipality. They found that many localized SME owners do not see a clear relative advantage in cloud migration because they lack basic knowledge about cloud capabilities, viewing it as an unnecessary expense rather than an efficiency driver.

Compatibility

Compatibility evaluates how well cloud solutions align with an organisation's existing business processes, operational workflows, values, and legacy technology architectures. Yaseen *et al.* (2023:320) establish that technological compatibility is the strongest positive predictor of long-term adoption intent among small business owners.

When cloud services integrate smoothly with everyday software tools (such as standard messaging or basic business systems), adoption barriers fall.

Conversely, Mkhize *et al.* (2025:online) point out that severe incompatibility arises when SMEs

rely on specialised on-premises legacy systems that cannot easily connect to cloud environments. This forces businesses into complex, costly modifications that often stall migration projects.

Complexity

Complexity represents the perceived difficulty of understanding, configuring, and operating cloud computing architectures. The literature shows complete agreement that complexity acts as a significant barrier to adoption.

While simple SaaS solutions present minimal complexity, configuring IaaS or PaaS environments demands advanced technical skills. Yaseen *et al.* (2023:320) emphasise that if IaaS and PaaS configurations are perceived as too complex, it acts as a direct inhibitor.

This issue is amplified for South African SMEs, which often operate without formal IT support or dedicated technical training.

Security Concerns

Security concerns focus on risks related to data breaches, unauthorised access, identity theft, and loss of control over sensitive business information. In global studies, security is often framed as a manageable technical challenge solved through encryption and multi-factor authentication.

However, Jayeola *et al.* (2022:228) show that within smaller enterprises, security is frequently viewed as an absolute inhibitor.

This perspective stems from a lack of trust in storing sensitive business data on shared, remote public servers managed by third parties. This trust deficit creates a paradox: while public cloud data centers are often more secure than basic on-premises servers, the perception of risk leads many SME owners to stick with vulnerable physical backup systems.

Data Privacy and POPIA Compliance

In the South African context, data privacy has evolved from a standard technical concern into a complex legal requirement with the implementation of the Protection of Personal Information Act (POPIA), which came into full effect in July 2021. POPIA mandates that enterprises implement strict protections when handling, processing, and transferring personal data, including restrictions on sending data across national borders.

Mokwena and Sithole (2024:online) identify POPIA compliance as a distinct technology-context barrier unique to South Africa, a finding not replicated at equivalent levels in comparable developing-economy studies from Jordan or Malaysia.

Because many global cloud hyperscalers use international data centres, SME owners often worry about cross-border compliance violations. Lacking internal legal or advanced technical expertise, many South African small businesses choose to avoid cloud migration entirely to prevent severe regulatory penalties.

4.6 Organisational Factors Influencing Cloud Adoption

Top Management Support

Top management support refers to the degree to which senior executives or small business owners understand, actively promote, and fund technology innovations. Within the SME sector, where organisational structures are often flat and centralised, the personal attitude of the business owner is a critical factor shaping firm behaviour.

The literature consistently shows that without strong management backing, cloud adoption projects rarely succeed.

Abu Bakar *et al.* (2024:556) note that when an owner values digital transformation, they can quickly overcome internal resistance and align resources. Conversely, if leadership is risk-averse or lacks digital literacy, the enterprise typically remains frozen in legacy operational models, even when employees push for cloud solutions.

Organisational Readiness and Financial Resources

Organisational readiness examines whether a company has the financial capital and physical resources needed to adopt cloud systems. A common point of debate in current research involves the true cost of cloud migration.

Proponents often argue that cloud computing lowers expenses by turning large capital investments into predictable operating costs. However, McKinsey and Company (2024:online) counter that capital constraints further prevent South African SMEs from absorbing migration costs, even where cloud represents a long-term saving over on-premises alternatives.

SMEs must pay for internet upgrades, data migration pipelines, dual-system running costs during migration, and consultant fees. For resource-constrained South African small businesses, these initial costs can create a significant financial barrier.

Digital Skills and IT Expertise

The successful adoption of advanced technology depends heavily on human capital capability. Mkhize *et al.* (2025:online) establish through a systematic review that inadequate digital skills and the absence of dedicated IT personnel are the foremost organisational barriers in sub-Saharan Africa.

Unlike large enterprises that maintain internal IT departments, typical South African SMEs rely on self-taught owners or a small staff with limited technical training.

This lack of internal IT expertise leaves businesses poorly equipped to evaluate cloud service providers, handle complex data migrations, or troubleshoot network configuration issues. Consequently, this skills gap increases reliance on expensive external consultants, raising the total cost of ownership and further discouraging small business owners from choosing cloud solutions.

4.7 Environmental Factors Influencing Cloud Adoption

Competitive Pressure

Competitive pressure represents the degree of influence an enterprise feels from competitors within its local market. Jayeola *et al.* (2022:233) demonstrate that competitive pressure from digitally-enabled rivals is a primary adoption motivator in SME-dense markets.

As larger corporations and digitally native start-ups utilise cloud systems to deliver faster

service and better customer experiences, traditional SMEs face strong pressure to adapt. Firms that fail to match the technological capabilities of their rivals risk losing market share. However, this pressure varies by location; while SMEs in urban hubs face intense competition from digital platforms, businesses in remote provinces often operate in less tech-dense local markets, which can reduce the perceived urgency to adopt cloud solutions.

Government and Vendor Support

The wider institutional environment is shaped by the level of support provided by state departments and technology vendors. Global research often highlights the positive impact of targeted government grants and digital enablement programs.

In South Africa, however, there is a clear gap between policy intentions and actual business support.

While the Department of Small Business Development has drafted policies targeting SME modernisation, actual ground-level support remains limited. Furthermore, Jayeola *et al.* (2022:237) and Mokwena and Sithole (2024:online) note that limited local vendor support for SMEs leaves them with minimal technical guidance, forcing small businesses to navigate complex legal compliance and migration challenges entirely on their own.

Infrastructure Constraints: Load Shedding and Connectivity

The foundational requirement for any cloud-based architecture is reliable, high-speed internet connectivity. This reliance exposes a critical difference between global and localised research. While international studies treat basic utility infrastructure as a given, South African researchers must address severe infrastructural deficits.

Adane (2021:online) and Mokwena and Sithole (2024:online) emphasize that persistent electricity supply disruptions undermine the reliable connectivity that cloud services require. When power cuts take down local cellular towers and business routers, cloud-hosted applications become completely inaccessible. This dynamic creates operational vulnerabilities, forcing many business owners to maintain expensive on-premises solutions, neutralizing the cost benefits of cloud models and slowing the pace of digital transformation.

4.8 Critical Synthesis of Literature

A synthesis of global and localised literature reveals that cloud computing adoption among SMEs is not shaped by isolated technical choices, but by a complex mix of technological, organisational, and environmental factors. While technological advantages like scalability and flexibility drive initial interest, internal organisational challenges—such as low digital skills and limited capital—often block actual migration.

This tension is further complicated by external environmental pressures. For instance, a small business may face strong competitive pressure to adopt agile cloud systems, yet find itself limited by localised power grid challenges and the legal complexities of POPIA compliance. Current research also highlights key contradictions between international studies and the realities on the ground in South Africa. Global models typically assume stable power grids and reliable internet access, framing cloud adoption as a strategic tool to optimize business processes.

In South Africa, however, infrastructure challenges like load shedding reshape the decision-making process. Cloud adoption is often viewed through the lens of business continuity, requiring specialised hybrid or offline-capable setups to handle frequent utility disruptions.

Furthermore, data protection laws create different challenges; where foreign firms might navigate mature, well-supported compliance frameworks, South African SMEs often view POPIA as a confusing legal hurdle that lacks clear, practical guidance for small businesses. To provide a comprehensive overview of how these dynamics have been analysed in prior scholarship, the table below synthesises key studies evaluating cloud adoption via the TOE framework:

Author(s) & Year	Country of Study	Research Methodology	Main Findings & Key Factors Identified	Aligned TOE Dimensions
Santos et al. (2023)	Portugal	Quantitative (Survey)	Relative advantage and compatibility strongly predict adoption; cloud integration directly improves overall MSME operational performance.	Technological, Organisational
Abu Bakar et al. (2024)	Malaysia	Quantitative (Survey)	Competitive pressure and top management support are the most influential drivers of cloud adoption intent within manufacturing SMEs.	Organisational, Environmental

Yaseen et al. (2023)	Jordan	Quantitative (Survey)	Technological compatibility is a primary driver, whereas high system complexity acts as a dominant inhibitor for SME owners.	Technological
Mokwena & Sithole (2024)	South Africa	Quantitative (Survey)	High costs, poor technical knowledge, and POPIA data privacy concerns act as severe adoption barriers within localized municipalities.	Technological, Environmental
Mkhize et al. (2025)	Sub-Saharan Africa	Systematic Review	Inadequate internal digital skills, a shortage of IT personnel, and high data costs represent the most critical barriers across the subcontinent.	Organisational

4.9 Research Gap

A deep engagement with contemporary information systems literature reveals a significant empirical gap. While the TOE framework has been widely applied globally, the vast majority of existing studies focus on high-income, developed economies or stable developing nations, such as Malaysia and Jordan. These models often assume a continuous supply of electricity and straightforward compliance pathways, which does not match the operational reality in

South Africa.

Furthermore, current South African scholarship frequently suffers from fragmentation. Existing localised studies often examine specific adoption barriers in isolation—focusing exclusively on infrastructure deficits or analyzing data protection laws without a clear theoretical framework. To date, no published study has applied the complete, integrated TOE framework to evaluate the combined technological, organisational, and environmental challenges facing South African SMEs.

Crucially, there is a clear shortage of research aimed at proposing a contextualised, practical adoption framework tailored specifically to the South African SME landscape.

Existing frameworks tend to offer broad, generic guidance that ignores the stark operational differences between well-resourced urban corporations and resource-constrained small businesses.

This study addresses these gaps by using a structured, qualitative systematic literature review to examine how internal resource constraints, infrastructure challenges like load shedding, and POPIA data compliance requirements interact.

By analysing these relationships through an integrated TOE lens, this research develops a contextualised adoption framework. This framework provides a practical roadmap for small business owners, technology vendors, and policymakers seeking to accelerate sustainable digital transformation across South Africa.

References

- Abu Bakar, M.R., Mat Razali, N., Ishak, K.K., Ismail, M.N. and Tengku Sembok, T.M. (2024). Adoption of Industry 4.0 with cloud computing as a mediator: evaluation using TOE framework for SMEs. *JOIV: International Journal on Informatics Visualization*, 8(2), pp. 554–563. <https://doi.org/10.62527/joiv.8.2.2205>
- Adane, M. (2021). Cloud computing adoption: opportunities and challenges for small, medium and micro enterprises in South Africa. *arXiv preprint, arXiv:2108.10079*. Available at: <https://arxiv.org/abs/2108.10079> [Accessed 12 April 2026].
- Ahmad, C.S., Mahomed, A.S.B. and Hashim, H. (2025). Cloud computing adoption in SMEs: exploring IaaS, PaaS and SaaS through a bibliometric study. *International Journal of Advanced Research in Science, Communication and Technology*, 15(1). Available at: <https://www.researchgate.net/publication/388409094> [Accessed 14 April 2026].
- Department of Small Business Development (DSBD). (2023). *Draft South African small, micro, and medium enterprises policy*. Pretoria: Government of South Africa.
- FinMark Trust. (2024). *FinScope MSME South Africa 2024*. Johannesburg: FinMark Trust. Available at: <https://finmark.org.za/knowledge-hub/articles/finscope-msme-south-africa-2024-key-findings-highlight-urgent-need-for-informal-sector-support> [Accessed 12 April 2026].
- Ilahi, S., Mohammed, R., Souissi, A., Hamati, M.Y.N. and Smida, R. (2024). Small and medium enterprises (SMEs): key factors affecting the adoption of cloud computing. *Saudi Journal of Humanities and Social Sciences*, 9(2), pp. 55–62. Available at: https://saudijournals.com/media/articles/SJHSS_92_55-62.pdf [Accessed 12 April 2026].
- IJSRA. (2024). Reviewing the impact of cloud computing on small and medium

enterprises in Africa. *International Journal of Science and Research Archive*, 11(01), pp. 1444–1451. Available at: <https://ijsra.net/sites/default/files/IJSRA-2024-0236.pdf> [Accessed 12 April 2026].

- Jayeola, O., Sidek, S., Abd Rahman, A., Mahomed, A.S.B. and Hu, J. (2022). Cloud computing adoption in small and medium enterprises (SMEs): a systematic literature review and directions for future research. *International Journal of Business and Society*, 23(1), pp. 226–243. <https://doi.org/10.33736/ijbs.4610.2022>
- McKinsey and Company. (2024). Africa's leap ahead into cloud: opportunities and barriers. *McKinsey Digital*, January. Available at: <https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/africas-leap-ahead-into-cloud-opportunities-and-barriers> [Accessed 12 April 2026].
- Mell, P. and Grance, T. (2011). *The NIST definition of cloud computing*. NIST Special Publication 800-145. Gaithersburg: National Institute of Standards and Technology.
- Mkhize, A., Mokhothu, K.D., Tshikhotho, M. and Thango, B.A. (2025). Evaluating the impact of cloud computing on SME performance: a systematic review. *Businesses*, 5(2), p. 23. <https://doi.org/10.3390/businesses5020023>
- Mokwena, L. and Sithole, B. (2024). Barriers of cloud computing adoption by SMEs in Polokwane Municipality. *Propulsion Technology Journal*, 45(2). Available at: <https://www.propulsionejournal.com/index.php/journal/article/download/6674/4364/11469> [Accessed 12 April 2026].
- Ngo Ndjama, R. and Van der Westhuizen, J. (2024). The role of small, medium, and micro enterprises in contributing to the socioeconomic development of South Africa. *International Journal of Research in Business and Social Science*, 13(7), pp. 606–619.
- Santos, A., Martins, J., Pestana, P., Gonçalves, R., São Mamede, H. and Branco, F. (2023). The antecedents of cloud computing adoption and its consequences for MSMEs' performance: a model based on the Technology-Organization-Environment (TOE) framework. *Cogent Business and Management*, 10(2), article 2220190. <https://doi.org/10.1080/23311975.2023.2220190>
- Statistics South Africa (Stats SA). (2024). *Gross domestic product: fourth quarter 2024*. Pretoria: Statistics South Africa. Available at: <https://www.statssa.gov.za/publications/P0441/P04414thQuarter2024.pdf> [Accessed 12 April 2026].
- Tornatzky, L.G. and Fleischer, M. (1990). *The processes of technological innovation*. Lexington, MA: Lexington Books.
- Yaseen, H., Al-Adwan, A.S., Nofal, M., Hmoud, H. and Abujassar, R.S. (2023). Factors influencing cloud computing adoption among SMEs: the Jordanian context. *Information Development*, 39(2), pp. 317–332. <https://doi.org/10.1177/026666669211047916>